PharmaChain Insights
Data-Driven Analytics for Pharma Supply Chain Operations

Batch & Real-Time Data Platform using AWS & Big Data

BUSINESS CONTEXT

- Pharmaceutical supply chains are complex and highly regulated
- Data is generated across manufacturing, quality labs, distributors, hospitals, and sensors
- Decisions are time-sensitive and compliance-critical

- In pharma, delayed or wrong decisions don’t just cost money — they create regulatory and safety risks.

CURRENT BUSINESS PROBLEMS

- Manufacturing, QC, inventory, and sales data exist in silos
- Cold-chain violations are detected too late
- Stock-outs are discovered after they impact patients
- Recall impact analysis is manual and slow
- Reporting is reactive instead of proactive

WHAT THE BUSINESS NEEDS

- A single source of truth across the supply chain
- Near real-time visibility into risks
- Automated, reliable analytics
- Scalable architecture as data grows
- Audit-ready historical data

SOLUTION OVERVIEW

- Unified batch + streaming data platform
- Centralized data lake on Amazon S3
- Real-time event processing for critical data
- Analytics using Athena & QuickSight

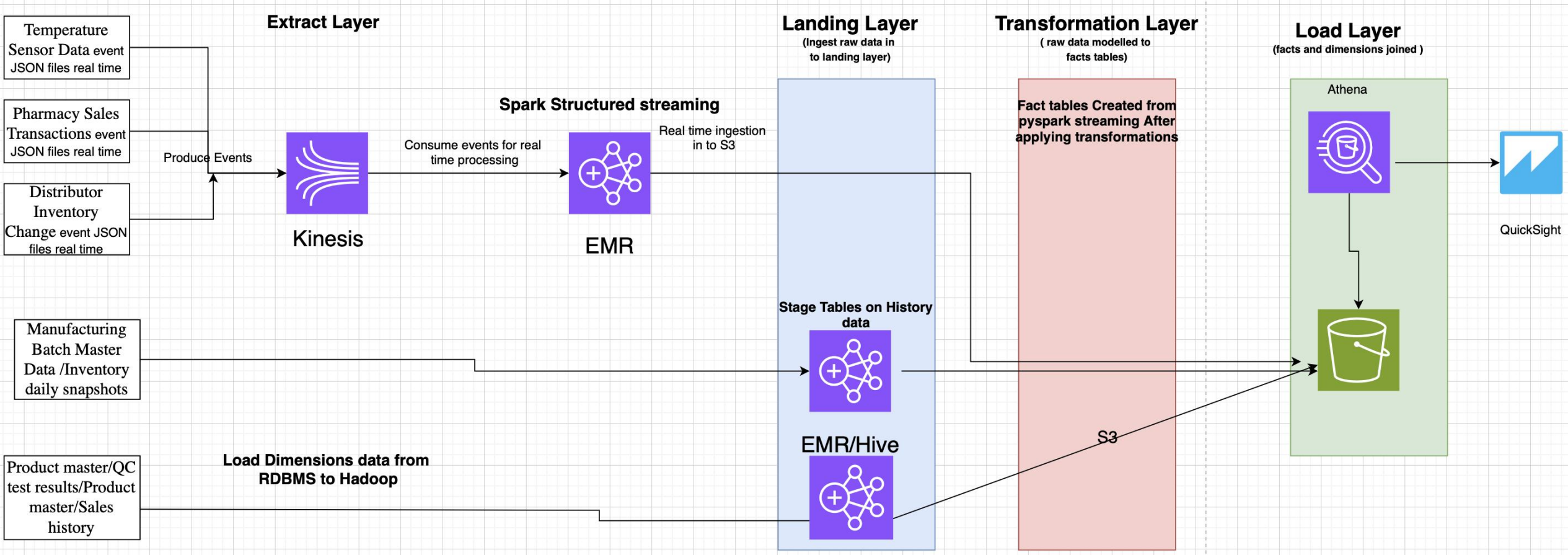
HIGH-LEVEL ARCHITECTURE

HIGH-LEVEL ARCHITECTURE (OVERVIEW)

- Extract: Manufacturing systems, QC labs, distributors, hospitals, sensors
- Ingest: Kinesis data streams for streaming, Hadoop/hive/Spark for batch
- Store: Amazon S3 (Raw, Cleansed, Curated)
- Process: Spark on EMR (batch & streaming)
- Analyze: Athena
- Visualize: QuickSight

- This architecture separates fast operational data from stable historical data and brings them together for decision-making.

Pharma Data Engineering Platform
Analytics Engine behind Pharma supply chain



Sample Realtime Queries

Query 1: Which high-value batches have crossed temperature limits in the last 1 hour and are nearing expiry?"

Query 2:Which regions will run out of stock based on current sales velocity and available inventory?

Query 3:How did a recall affect sales in the last 2 hours compared to normal?

Query 4: Which distributors have repeated temperature violations across multiple batches?

Query 5:Are any hospitals buying unusually high quantities compared to historical patterns?

Query 6:Are all active batches compliant with QC and temperature requirements?

Datasets

- STREAMING DATASETS(JSON)
  - Cold-chain temperature sensor events
  - Hospital / pharmacy sales transactions
  - Distributor inventory updates
- HISTORICAL (BATCH) DATASETS(JSON/RDS)
  - Manufacturing batch master
  - QC test results
  - Product master
  - Distributor & warehouse master
  - Daily inventory snapshots
  - Sales history
  - Recall history

KEY BUSINESS QUESTIONS THIS PLATFORM ANSWERS

Questions

- Which batches are at immediate risk?
- Where will stock-outs occur next?
- Which distributors cause repeated violations?
- How do recalls impact sales in real time?
- Are we audit-ready at any point in time?
- Cold chain compliance — which batches had temperature breaches in the last 24 hrs?
- QC risk on live orders — sales of batches that later failed QC and are still in transit
- Stock coverage — how many days of stock does each distributor have based on recent sales velocity?

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- Query 6:Are all active batches compliant with QC and temperature requirements?